



Molecular phylogenetics, phylogenomics, and phylogeography

Phylogeny, divergence times, and biogeography of the subfamily Tagiadinae (Lepidoptera: Hesperidae) based on molecular data and morphological characters

Xiangyu Hao¹, Yue Pan¹, Hideyuki Chiba^{2,✉}, and Xiangqun Yuan^{1,*,}

¹Key Laboratory of Plant Protection Resources and Pest Management, Ministry of Education, Entomological Museum, College of Plant Protection, Northwest A&F University, Yangling, China

²B.P. Bishop Museum, Honolulu, HI, USA

*Corresponding author. Xiangqun Yuan, Key Laboratory of Plant Protection Resources and Pest Management, Ministry of Education, Entomological Museum, College of Plant Protection, Northwest A&F University, Yangling 712100, China (Email: yuanxq@nwsuaf.edu.cn).

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The skipper subfamily Tagiadinae has been a key group in taxonomic debates and phylogenetic inconsistencies due to limited taxon sampling and morphological evidence. In this study, we aimed to resolve intra-subfamilial relationships within Tagiadinae using 3 datasets: (i) a molecular dataset (3 genes *COI*, *COII*, *EF-1α*) including 92 species from 30 genera across all the 3 tribes of Tagiadinae, (ii) a morphological dataset (62 morphological or biological traits) comprising 50 species from 20 genera, and (iii) the combined dataset, representing the intersection of the first 2, containing 46 species from 20 genera. Both the molecular tree and the combined molecular-morphological tree supported the monophyly of Tagiadinae and its internal tribes, and the tribe Netrocorynini was consistently recovered as the sister group of the other Tagiadinae species. In addition, divergence time estimation suggested a crown age of approximately 37 million years for Tagiadinae, with the 2 tribes Celaenorrhinini and Tagiadini having diverged approximately 36 million years ago. The morphological and molecular evidence presented in this study contributes to a more robust understanding of the evolutionary framework of Tagiadinae. Our findings challenge some previous phylogenetic hypotheses regarding the basal position of Netrocorynini within Tagiadinae and provide valuable insights into the taxonomy and phylogeny of skipper butterflies.

Keywords: skipper, phylogenetic analysis, morphology.

Introduction

In recent years, the field of phylogenetics has seen a growing reliance on molecular data, often at the expense of traditional morphological evidence. While molecular data provide many advantages, such as high resolution and broad applicability, morphological data remain crucial, particularly for understanding the evolutionary relationships of fossil species and linking them to living organisms (Wiens 2004, Giribet 2015). This dual approach is essential not only for reconstructing the evolutionary history of specific groups but also for broader insights into the processes that drive biodiversity and adaptation.

The family Hesperidae, commonly known as skipper butterflies, was established by Latreille in 1809. The globally distributed

skippers consist of about 567 genera and over 4,000 species and account for one-fifth of the world's butterflies (Warren et al. 2008). Current molecular systematics based on low-coverage genomes established the phylogenetic framework of the 13 subfamilies of Hesperidae (Li et al. 2019, Zhang et al. 2019, 2020, Kawahara et al. 2023). Among them, Tagiadinae was promoted to the subfamily rank (Li et al. 2019), containing 2 traditional genus groups (*Celaenorrhinus* group and *Tagiades* group) within Pyrginae *sensu lato* Evans (1949). Current phylogenetic consensus recognizes the tribe-level status of Tagiadini, Celaenorrhinini (Evans 1949, 1951, Warren et al. 2008, 2009), and newly established Netrocorynini, which comprises 3 genera (*Chaetocneme*, *Exometoeca*, and

Netrocoryne), was separated from Tagiadini (Li et al. 2019). There are 35 genera and 311 described species in Tagiadae in the world, mainly distributed in the Oriental Region and the Afrotropical Region (Evans 1949, Devyatkin 1996, Warren et al. 2009, Cock and Congdon 2012, Yuan et al. 2015, Huang et al. 2019).

The intra-subfamilial relationships of Tagiadae were quite disputed at times. Sahoo et al. (2016) and Sahoo et al. (2017) sampled 7 genera and 11 genera (both including *Netrocoryne*) of Tagiadini, respectively, but obtained 2 distinct topologies. The most significant difference between the 2 studies is that the clade consisting of (*Eagris* + *Tagiades*) is at the tip (Sahoo et al. 2016) or the basal position (Sahoo et al. 2017) of Tagiadini. The results of Toussaint et al. (2018) which used Anchored Hybrid Enrichment methods were also different from previous studies, but mainly due to the different taxon sampling. Li et al. (2019) made major revisions to skipper classification, including the establishment of the subfamily Tagiadae and tribe Netrocorynini mentioned above. In addition, mitochondrial genomic data revealed the relationships within the tribe Tagiadini as (*Tagiades* + ((*Satarupa* + *Mooreana*) + (*Capila* + ((*Odontoptilum* + *Abreximorpha*) + (*Ctenoptilum* + *Gerosis*)))) (Xiao et al. 2022).

Given long-standing phylogenetic inconsistencies and the lack of comprehensive morphological studies, in this study, we compiled a more comprehensive molecular dataset (3 genes *COI*, *COII*, *EF-1α*) for phylogenetic reconstruction and divergence time estimation, aiming to provide a robust time-calibrated evolutionary framework. In addition, a morphological dataset (62 morphological or biological traits), as well as a molecular-morphological combined dataset based on the intersection of the former 2, were also utilized for phylogenetic reconstruction to provide additional morphological evidence.

Materials and Methods

Taxon Sampling

To reconstruct the phylogeny of Tagiadae, we utilized 3 distinct datasets: (i) a molecular dataset comprising 92 species from 30 genera across all the 3 tribes within the subfamily Tagiadae, (ii) a morphological dataset consisting of 50 species from 20 genera, selected based on the availability of specimens with sufficient morphological data for analysis, and sampling aimed to include representatives from across the subfamily, and (iii) a molecular-morphological combined dataset including 46 species from 20 genera, representing the intersection of taxon sampling between the 2 former datasets. Three representative species from 3 hesperiid subfamilies Barcinae, Hesperinae, and Heteropterinae, were selected as outgroups. Details are provided in Table 1.

Molecular Data

All materials were preserved in absolute ethanol after collecting and stored at -20 °C environment in the Entomological Museum of Northwest A&F University, Yangling, China. Genomic DNA was isolated from thoracic or leg tissue using the Qiagen QIAamp DNA Micro Kit (Cat No. 56304). The genes *COI*, *COII*, and *EF-1α* were amplified by PCRs using previously reported primers (Folmer et al. 1994, Cho et al. 1995, Monteiro and Pierce 2001, Hebert et al. 2004, Wan et al. 2013) (Table 2). PCR products were sequenced using Sanger sequencing (Tsingke Biotechnology, Beijing, China). Next, we integrated available sequences and additionally assembled the Sequence Read Archive data in NCBI using Geneious R9 (Biomatters, Auckland, New Zealand) to supplement the molecular data matrix (Table 1).

The open reading frames of 3 genes were manually inspected using MEGA7. Sequence alignment was conducted using MAFFT with L-INS-i strategy (Katoh and Standley 2013). Individual genes were concatenated using PhyloSuite (Xiang et al. 2023). The optimal partition schemes and best nucleotide substitution models of maximum likelihood (ML) and Bayesian inference (BI) methods were predicted using PartitionFinder2 (Lanfear et al. 2017). We partitioned the concatenated dataset based on gene position and further subdivided each gene by codon position. Phylogenetic relationships were reconstructed using IQ-TREE (Nguyen et al. 2015) and MrBayes (Ronquist et al. 2012) based on ML and BI methods, respectively. ML tree was inferred from the Ultrafast bootstrap (UFB) algorithm, and the support value of each node was evaluated by 10,000 UFB replicates. For BI method, 2 independent runs of 2×10^7 generations were implemented with 4 independent Markov Chain Monte Carlo (MCMC) chains. A consensus tree was obtained when the average standard deviation of split frequencies <0.01, and the initial 25% trees were discarded as burn-in in each MCMC run. Support values of BI analyses were estimated by the Bayesian posterior probability.

Morphology

A total of 87 specimens were dissected. The abdominal region was treated with a 10% sodium hydroxide solution, followed by boiling and removal of impurities to obtain clear genital structures. For slide preparation, vibrant and finely detailed genitalia were transferred to glass slides using glycerol and labeled accordingly. Finally, images were captured using a Nikon Cool Pix 5000 digital camera for documentation.

In this study, a total of 62 (including 45 external morphological characters, 14 female and male genital traits, and 3 biological traits) characters were selected from the subfamily Tagiadae according to previous studies (Wahlberg et al. 2005, Warren 2006, Warren et al. 2009). Binary coding was employed for dichotomous features, while additive encoding was used for multistate traits.

Characters and states for cladistic analysis are categorized as follows: Antenna (01–04), Head (05–11), Thorax (12–13), Leg (14–20), Wing (21–42), Abdomen (43–45), Genitalia (46–59), and biological traits (60–62).

01. Antenna shape: Straight, lacking terminal bulge (0); Midsection of the shaft curved with a spear-like terminal bulge (1); Midsection of the shaft curved with a short, thick, blunt terminal (2).
02. Antenna base distance: Less than 2 times the width of the scape (0); Greater than 2 times the width of the scape (1).
03. Antennal length relative to forewing anterior margin: Less than half the length of forewing (0); Equal to or longer than half the length of forewing (1).
04. Antenna color: Uniform (0); Variegated (1).
05. Front chaetosema: Absent (0); Present (1).
06. Eyelashes: Absent (0); Present (1).
07. Labial palpus segment II: Narrow (0); Thicker than segment III (1); Very wide, about twice as thick as segment III (2).
08. Angle between labial palpus segments II and III: Obtuse angle (0); Nearly right angle (1).
09. Labial palpus segment III shape: 1-3 times longer than wide, blunt at apex (0); More than 3 times longer than wide, blunt at apex (1); More than 4 times longer than wide (2).
10. Maxillary palpus: Absent (0); Present (1).
11. Labial palpus color compared to body color: Clearly different (0); Same (1).

Table 1. Taxon sampling, collection information and data used in this study

Tribe	Species	Collection information	Data
Celaenorrhinini	<i>Aurivittia aurivittata</i>		Morphology*, COI (NC_083975.1), COII (NC_083975.1)
	<i>Aurivittia vietnamicus</i>	Yunnan, China 2022.8	Morphology*, COI*, COII*, EF-1 α *
	<i>Celaenorrhinus maculosus</i>	Fujian, China 2021.8	Morphology*, COI*, COII*, EF-1 α *
	<i>Celaenorrhinus consanguineus</i>	Sichuan, China 2021.8	Morphology*, COI*, COII*, EF-1 α *
	<i>Celaenorrhinus kiku</i>	Guangdong, China 2021.8	Morphology*, COI*
	<i>Celaenorrhinus aspersus</i>	Sichuan, China 2020.8	Morphology*, COI*, COII*, EF-1 α *
	<i>Celaenorrhinus pulomaya</i>		Morphology*
	<i>Celaenorrhinus patula</i>		Morphology*, COI, COII, EF-1 α (SRR9330396)
	<i>Celaenorrhinus pyrrha</i>		COI (MT997276.1),
	<i>Celaenorrhinus leucocera</i>		COI (OR195690.1)
	<i>Celaenorrhinus tibetana</i>		COI (PQ252722.1)
	<i>Celaenorrhinus eligius</i>		COI (ON436817.1), EF-1 α (OK737228.1)
	<i>Celaenorrhinus qianbanae</i>		COI (OR195688.1)
	<i>Celaenorrhinus macrostictus</i>		COI (ON437147.1), EF-1 α (OK737603.1)
	<i>Celaenorrhinus victor</i>		COI (MZ501805.1), COII (MZ501805.1)
	<i>Apallaga homeyeri</i>		COI (ON436374.1), EF-1 α (OK736800.1)
	<i>Pseudocoladenia dan</i>	Sichuan, China 2020.8	Morphology*, COI*, COII*, EF-1 α *
	<i>Pseudocoladenia dea</i>	Sichuan, China 2009.6	Morphology*, COI*, EF-1 α *, COII (MZ821244.1)
	<i>Pseudocoladenia festa</i>		Morphology*, COI (NC_079686.1), COII (NC_079686.1)
	<i>Pseudocoladenia decora</i>		COI (OQ152244.1)
	<i>Pseudocoladenia eacus</i>		COI (OQ152250.1)
	<i>Pseudocoladenia yunnana</i>		COI (OQ152287.1)
	<i>Pseudocoladenia celebica</i>		COI (OQ152229.1)
	<i>Pseudocoladenia fatua</i>		COI (OQ152268.1)
	<i>Pseudocoladenia fatih</i>		COI (OQ152265.1)
	<i>Pseudocoladenia sadakoe</i>		COI (OQ152284.1)
	<i>Pseudocoladenia fulvescens</i>		COI (OQ152283.1)
	<i>Sarangesa dasahara</i>	Yunnan, China 2017.8	Morphology*, COI*, COII*, EF-1 α *
	<i>Triskelionia tricerata</i>		COI (ON436001.1), EF-1 α (OK736447.1)
Tagiadini	<i>Abantis leucogaster</i>	Indonesia 2019.11	COI*, COII*
	<i>Abantis efulensis</i>	Cameroon 2017.8	Morphology*, COI*, COII*
	<i>Abantis ja usheri</i>		COI (ON435720.1), EF-1 α (OK736189.1)
	<i>Abraximorpha davidii</i>	Zhejiang, China 2016.6	Morphology*, COI*, COII*, EF-1 α *
	<i>Abraximorpha esta</i>	Vietnam 2020.3	Morphology*, COI*, COII*, EF-1 α *
	<i>Capila translucida</i>	Hainan, China 2018.4	Morphology*, COI*, COII*, EF-1 α *
	<i>Capila hainana</i>	Vietnam 2021.4	COI*, COII*, EF-1 α *
	<i>Capila jayadeva</i>	Vietnam 2017.4	Morphology*, COI*, COII*, EF-1 α (OK736837.1)
	<i>Capila pennicillatum</i>	Vietnam 2019.5	Morphology*, COI*, COII*, EF-1 α *
	<i>Capila neolineata</i>	Hunan, China 2020.7	Morphology*, COI*, COII*, EF-1 α *
	<i>Capila lineata</i>		COI (OP861540.1)
	<i>Capila omeia</i>		COI (OP825043.1)
	<i>Capila phanaeus phanaeus</i>		COI (ON435773.1), EF-1 α (OK736237.1)
	<i>Caprona agama</i>	Yunnan, China 2022.8	Morphology*
	<i>Caprona alida</i>		Morphology*, COI*, COII*, EF-1 α *
	<i>Coladenia hoenei</i>	Shanxi, China 2020.6	Morphology*, COI*, COII*, EF-1 α *
	<i>Coladenia vitrea</i>	Shanxi, China 2007.5	Morphology*, COI*, EF-1 α *, COII (MZ821220.1)

Table 1. Continued

Tribe	Species	Collection information	Data
	<i>Coladenia laxmi</i>	Guangdong, China 2020.7	Morphology*, COI*, COII*, EF-1 α *
	<i>Coladenia agnioides</i>		COI (NC_087893.1), COII (NC_087893.1)
	<i>Coladenia maeniata</i>		COI (NC_079683.1), COII (NC_079683.1)
	<i>Coladenia pinsbukana zhui</i>		COI (OR500957.1)
	<i>Ctenoptilum vasava</i>	Thailand 2012.3	Morphology*, COI*, COII*, EF-1 α *
	<i>Darpa hanria</i>		Morphology*
	<i>Darpa pteria</i>	Philippines 2019.10	Morphology*, COI*, COII*
	<i>Darpa inopinata</i>		COI (NC_079684.1), COII (NC_079684.1)
	<i>Gerosis sinica</i>	Guangdong, China 2020.7	Morphology*, COI*, COII*, EF-1 α *
	<i>Gerosis phisara</i>	Zhejiang, China 2017.7	Morphology*, COI*, COII*, EF-1 α *
	<i>Gerosis bhagava</i>	Burma 2010.10	COI*, COII*, EF-1 α *
	<i>Leucochitonea levubu</i>		COI (ON436881.1), EF-1 α (OK737291.1)
	<i>Mooreana trichoneura</i>	Yunnan, China 2017.7	Morphology*, COI*, COII*, EF-1 α *
	<i>Mooreana princeps</i>		COI (ON435879.1), EF-1 α (OK736333.1)
	<i>Odina decoratus</i>		COI (PQ252760.1)
	<i>Odina hieroglyphica</i>		COI (ON435899.1), EF-1 α (OK736351.1)
	<i>Odontoptilum angulatum</i>	Yunnan, China 2022.8	Morphology*, COI*, COII*, EF-1 α *
	<i>Odontoptilum pygela</i>	Indonesia 2020.2	Morphology*, COI*, COII*, EF-1 α *
	<i>Pintara bowringi</i>	Hunan, China 2020.7	Morphology*, COI*, COII*, EF-1 α *
	<i>Pintara melli</i>	Hunan, China 2020.7	COI*, COII*, EF-1 α *
	<i>Pintara pinuilli</i>	Vietnam 2021.4	Morphology*, COI*, COII*, EF-1 α *
	<i>Pintara heringi heringi</i>		COI (OR606593.1)
	<i>Pintara pieridoides</i>		COI (OR606591.1)
	<i>Procampta rara</i>		COI (ON435939.1), EF-1 α (OK736391.1)
	<i>Satarupa monbeigi</i>	Jiangxi, China 2007.7	Morphology*, COI*, COII*, EF-1 α *
	<i>Satarupa nymphalis</i>	Zhejiang, China 2017.7	Morphology*, COI*, COII*, EF-1 α *
	<i>Satarupa valentini</i>		Morphology*, COI (MZ711376.1) COII (MZ821216.1)
	<i>Satarupa zulla</i>	Sichuan, China 2020.6	Morphology*, COI*, COII*, EF-1 α *
	<i>Satarupa gopala</i>		COI (ON435966.1), EF-1 α (OK736418.1)
	<i>Seseria dohertyi</i>	Yunnan, China 2022.8	Morphology*, COI*, COII*, EF-1 α *
	<i>Seseria sambara</i>	Xizang, China 2016.6	Morphology*, COI*, COII (MZ821242.1), EF-1 α *
	<i>Seseria strigata</i>		COI (PQ252783.1)
	<i>Tagiades litigiosa</i>	Hainan, China 2018.4	Morphology*
	<i>Tagiades gana</i>	Yunnan, China 2019.7	Morphology*, COI*, COII*, EF-1 α *
	<i>Tagiades menaka</i>	Hainan, China 2018.4	Morphology*, COI*, COII*, EF-1 α *
	<i>Tagiades cohaerens</i>	Japan 2019.5	Morphology*, COI*, COII*, EF-1 α *
	<i>Tagiades trebellius</i>	Japan 2019.6	Morphology*, COI*, COII*, EF-1 α *
	<i>Tagiades nestus</i>	Indonesia 2020.4	Morphology*, COI (HQ570876.1), COII*, EF-1 α *
	<i>Tagiades japetus</i>	Philippines 2019.7	Morphology*, COI*, COII*, EF-1 α (MN842168.1)
	<i>Tagiades tethys</i>		Morphology*, COI*, COII (KJ813807), EF-1 α (MN842159.1)
	<i>Tagiades vajuna</i>		COI (KX865091.1), COII (KX865091.1), EF-1 α *
	<i>Tagiades flesus</i>		COI (ON437267.1), EF-1 α (OK737711.1)
	<i>Tapena thwaitesi</i>	Philippines 2019.7	Morphology*, COI*, COII*, EF-1 α *
	<i>Calleagris jamesoni jamesoni</i>		COI (ON435767.1), EF-1 α (OK736231.1)
	<i>Eagris sabadius</i>		COI (ON435812.1), EF-1 α (OK736270.1)
	<i>Netrobalane canopus</i>		COI (ON435889.1), EF-1 α (OK736343.1)

Table 1. Continued

Tribe	Species	Collection information	Data
Netrocorynini	<i>Netrocoryne repanda</i>		Morphology, <i>COI</i> *, <i>COII</i> (MW288058.1), <i>EF-1α</i> (EU364136.1)
	<i>Exometoeca nycteris</i>		<i>COI</i> (KF392677.1), <i>EF-1α</i> (OK737791.1)
	<i>Chaetocneme critomedia</i>		<i>COI</i> (KF394440.1)
	<i>Chaetocneme helirius</i>		<i>COI</i> (ON435783.1)
Outgroups	<i>Barca bicolor</i>		Morphology*, <i>COI</i> *, <i>COII</i> (NC_039947.1), <i>EF-1α</i> (MZ165624.1)
	<i>Halpe nephele</i>	Fujian, China 2016.7	Morphology*, <i>COI</i> *, <i>COII</i> (MW288058.1), <i>EF-1α</i> *
	<i>Heteropterus morpheus</i>	Hebei, China 2022.5	Morphology*, <i>COI</i> *, <i>COII</i> (NC_028506), <i>EF-1α</i> *

Asterisks (*) represent sequences amplified by PCR in this study. All the accession numbers of available sequences in NCBI are provided.

Table 2. Primer sequences used in this study

Gene	Primer	Sequence (5'–3')	Reference
<i>cox11-1</i>	COI1490	GGTCAACAAATCATAAAGATATTG	Folmer et al. (1994)
	LEP-R2	CTTATATTATTTATTCGTGGGAAAGC	Hebert et al. (2004)
<i>cox11-2</i>	F1	ACKAATCATAAAGATATTGGAAC	Hebert et al. (2004)
	R1	AAAATWGCTAARTCAACWGAAG	
<i>cox12-1</i>	F2	GAAGAATTGTAGAAAATGGTKCWG	
	COI2198	TAAACTTCAGGGTGACCAAAAAAT	Folmer et al. (1994)
<i>cox12-2</i>	F2	GAAGAATTGTAGAAAATGGTKCWG	
	R2	CTTTCTTGAGAAATAARTTGRG	Hebert et al. (2004)
<i>cox2</i>	Pierre	AGAGCCTCTCCTTTAATAGAACA	Bogdanowicz et al. (1993)
	Eve	GAGACCATTACTTGCTTTCAGTCATCT	Bogdanowicz et al. (1993)
<i>cox21</i>	George	ATACCTCGACGTTATTGAGA	Bogdanowicz et al. (1993)
	Phyllis	GTAATAGCAGGTAARATAGTTCA	Monteiro and Pierce (2001)
<i>cox22</i>	Strom	TAATTTGAACTATYTTACCAGC	Monteiro and Pierce (2001)
	R2	TTACTTACTTTCAGCCATCT	
<i>EF-1α</i>	ELF2F	AAAATGCCCTGGTTCAAGGGA	Wan et al. (2013)
	efrcM4	ACAGCVACKGTYTGYCTCATRTC	Wan et al. (2013)
<i>EF-1α2</i>	F2	CGWGGYTAYGTYGCTGGTGA	
	R2	GCTACTTYTTGCCCTTRGTG	

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|---|---|
| <p>12. Patagia: Broader than prothorax, distinctly separated from humeral plate (0); Narrower than prothorax, fused with humeral plate (1).</p> <p>13. Male patagium: Absent (0); Composed of large oblique scales (1).</p> <p>14. Outer hair-pencil on male hindtibia: Absent (0); Present (1).</p> <p>15. Inner hair-pencil on male hindtibia: Absent (0); Present (1).</p> <p>16. Inner hair-pencil on male forecoxa: Absent (0); Present (1).</p> <p>17. Foretibia strigilis: Prominent and developed (0); Reduced or slender (1).</p> <p>18. Foretibia spur: Absent (0); Few (1).</p> <p>19. Pterothorax foretibia spur: Absent (0); Few (1).</p> <p>20. Two midsegment spurs of the hind tibia: Equal (0); Unequal (1).</p> <p>21. Humeral plate fringe: Shorter than humeral plate (0); Longer than humeral plate (1).</p> <p>22. Forewing M_2 vein position: Adjacent to M_3 vein (0); Intermediate between M_1 and M_3 veins (1); Adjacent to M_1 vein (2).</p> | <p>23. Length of forewing median cell compared to that of posterior margin: Shorter than posterior margin (0); Equal to or longer than posterior margin (1).</p> <p>24. Length of forewing median cell compared to forewing length: Less than 60% of forewing length (0); Equal to or longer than 60% of forewing length (1).</p> <p>25. Forewing median cell: Closed (0); Slightly open (1); Open (2).</p> <p>26. Forewing M_1-M_3: Not curved at M_2 base (0); Slightly curved (1).</p> <p>27. Distance between forewing R_5 + M_1 and R_5 + R_4: Distant (0); Equal (1); Close (2).</p> <p>28. Forewing 3A and 2A: Separated (0); Fused (1).</p> <p>29. Male forewing anterior margin fold: Absent (0); Present (1).</p> <p>30. Male forewing androconial patch: Absent (0); Present (1).</p> <p>31. Male forewing androconial mark on the underside: Absent (0); Present (1).</p> <p>32. Ventral margin fringed scales on forewing: Absent (0); Present (1).</p> <p>33. Hindwing median cell compared to hindwing length: Shorter than half of hindwing length (0); Equal to or longer than half of hindwing length (1).</p> |
|---|---|

34. Hindwing posterior margin: Smooth (0); Protruding at the end of R_s and M_3 veins (1); Protruding at the end of R_s and Cu_1 veins (2).
35. Hindwing anterior margin tuft (wing coupling structure): Absent in both females and males (0); Present in both females and males (1).
36. Glossy, short scales at R_s - Cu base on the dorsal side of hindwing: Absent (0); Present (1).
37. Hindwing shoulder: Thickened, thumb-like (0); No thickening at the margin or with short veins at the basal vein (1).
38. Hindwing median veins: 3 veins extending to wing margin (0); Weak M_2 vein (1); Only 2 median veins, lacking M_2 (2).
39. Hindwing M_3 and Cu_1 (based on median cell): Well-separated (0); Fused (1).
40. Hindwing humeral vein: Absent (0); Short (1).
41. Hindwing cell: Closed (0); Open, very small, horizontal or extending forward to base of R_s (1).
42. Elongated marginal seta at hindwing anal angle: Absent (0); Present (1).
43. First tergite of abdomen: Flat or slightly arched (0); Inclined to nearly vertical position (1).
44. Length of male abdomen: Shorter than or equal to hindwing posterior margin (0); Longer than hindwing posterior margin (1).
45. Anal seta on female eighth abdominal segment: Absent (0); Present (1).
46. Symmetry of female genitalia: Bilateral symmetry (0); Asymmetry (1).
47. Corpus bursae: Membranous (0); Ossified region short (1); Ossified region long (2).
48. Cercus spine at apex: Absent (0); Present (1).
49. Morphology of bursa copulatrix: Pouch-shaped (0); Midsection constricted, gourd-shaped (1).
50. Valvae symmetry: Symmetrical (0); Asymmetrical (1).
51. Valvae apex shape: Intact, without fissures (0); Bifurcated (1); Trifurcated (2).
52. Paramere: Absent (0); Present (1).
53. Uncus: Overall narrow, apex unsplit (0); Overall narrow, apex split (1); Overall wide, apex unsplit (2); Overall wide, apex split (3).
54. Gnathos: Fused (0); Separated (1).
55. Saccus: Short (0); Long (1).
56. Cornuti: Absent (0); Present (1).
57. Valva costa shape: Lamellar (0); Annular (1).
58. Aedeagus: Short and stout (0); Moderate in shape (1); Long and slender (2).
59. Ejaculatory duct connection position: Near the anterior end (0); Near the middle (1).
60. Adult resting posture: Wings erect (0); Wings flat or overlapping above abdomen (1).
61. Adult sunbathing posture: Forewings slanted, hindwings spread flat (0); Forewings and hindwings spread flat (1).
62. Larval host plant: Dicotyledonous (0); Monocotyledonous (1).

The phylogenetic tree for morphological data was constructed using maximum parsimony (MP) method. By utilizing the software Mesquite 3.10 (Maddison 2008), we conducted the entry of taxonomic characters and generated files in the “TNT” format. Subsequently, the phylogenetic tree was constructed using TNT 1.1 software (Goloboff et al. 2008) based on the New Technology search with the parameters: Tree fusing, the random seed was set to 1, Auto-constrain, Replace existing trees, Driven search: “Init. addseqs” was set to 5, Find min. length: 1 times. All characters were given equal weight.

The TNT software computed the Bootstrap values (BS) and Relative Bremer Support values (BSV) for each node. WinClada 1.0 software was enlisted to annotate alterations in features, whereby the synapomorphies were denoted by black circles, and nonhomologous characters were indicated by open circles.

Molecular-Morphological Phylogeny

Due to the limited taxon sampling of morphological data and given the extensive missing data for some specific taxa with only morphological data in the next molecular-morphological combined analysis, we constructed a new dataset that only including taxa with both molecular and morphological data. This dataset was used to infer phylogenetic relationships incorporating morphological evidence.

Phylogenetic tree was constructed was conducted using TNT software. The script “PhylogenomicSearch.run” (<https://github.com/atorresgalvis/TNT-scripts-for-phylogenomics/tree/main/Scripts>) was used to generate the strict consensus tree based on the New Technology method, and the parameter “Weighting against homoplasy” was set to “Implied weighting”. Next, the nodal support values were evaluated using the script “PhylogenomicSupport.run”. Support analyses were conducted by resampling individual sites (JS) and individual genes (JG) using the Jackknifing method with 100 replicates, complemented by Relative Bremer Support tests.

Divergence Time Estimation and Ancestral Region Reconstruction

The divergence times were estimated using MCMCtree program in the PAML package (Yang 2007). We first used baseml program to roughly estimate the overall substitution rate of the dataset based on the GTR + R model. Next, MCMCtree was used to estimate the gradient and Hessian of the likelihood values, branch lengths, and divergence times. Due to the lack of fossil data, we used 4 secondary calibrations (see Chazot et al. 2019 and its supplementary files) in the molecular dating analyses. Next, we reconstructed the ancestral biogeography using RASP (Yu et al. 2015) based on the DEC model. The distribution information of extant species was integrated from 2 resources: the Global Biodiversity Information Facility (<https://www.gbif.org/>) and WikiSpecies (<https://species.wikimedia.org>). The maximum number of ancestral areas of this program was set to 2, and the other parameters to default were allowed to default. All geographic information used for ancestral region reconstruction analysis is shown in Fig. 3.

Results

Molecular Phylogenetic Trees

Inclusive of 155 sequences encompassing 23 genera and 54 species, our sequences were determined through PCR experiments. Specifically, these sequences consist of 54 segments of 660 bp from the *COI* gene, 53 segments of 621 bp from the *COII* gene, and 48 segments of 642 bp from the *EF-1 α* gene. In addition, we expanded our taxon sampling by integrating available sequences in NCBI and eventually compiled a molecular dataset covering 92 species in 30 genera.

Both ML and BI analyses yielded nearly identical topologies, except for the intra-generic relationships within the genus *Seseria*. For simplicity and brevity, only one phylogenetic hypothesis (ML method) is presented here (Fig. 1A). The trees with detailed branch lengths and support values were provided in Supplementary Figs. S1 and S2.

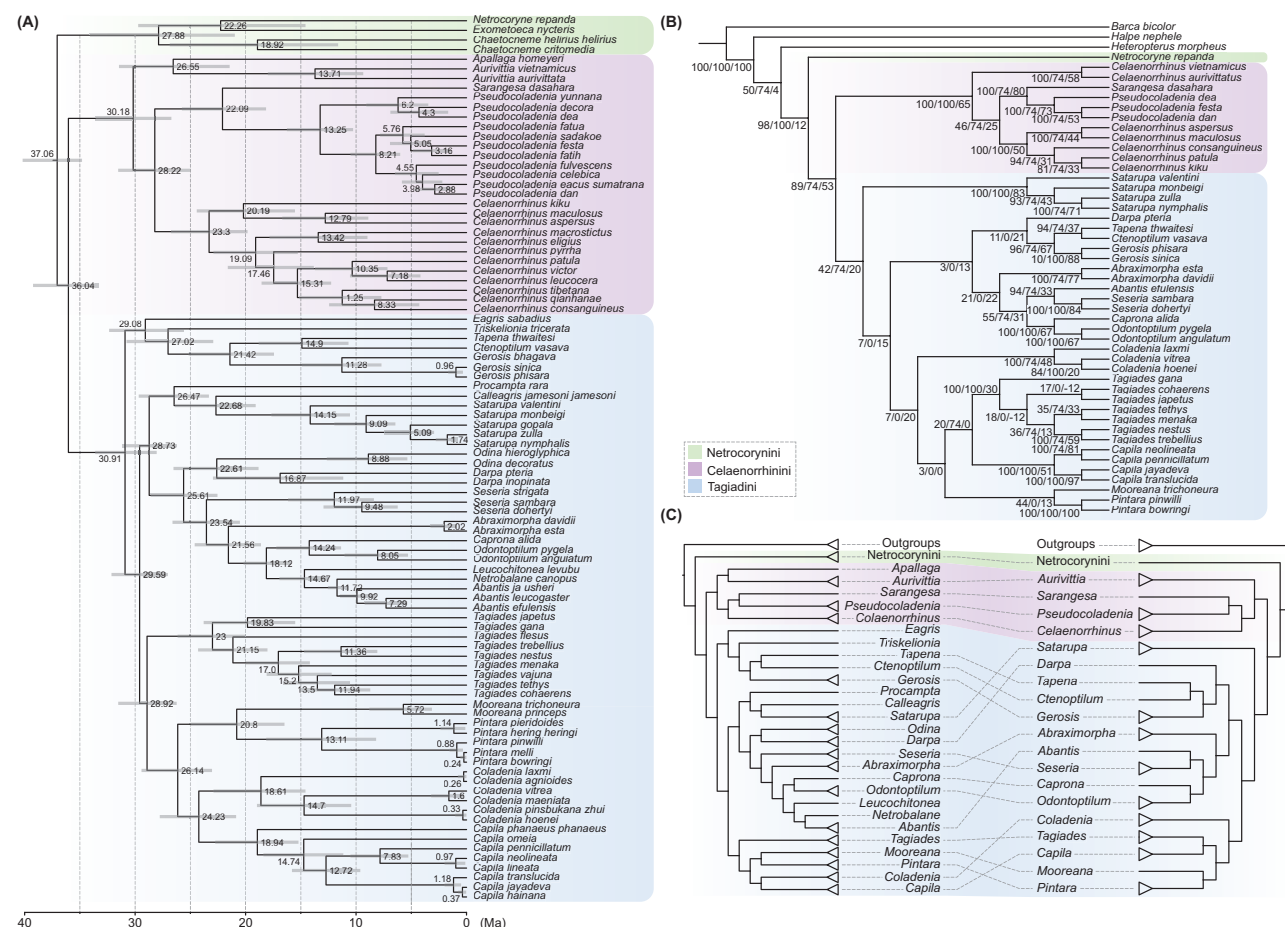


Fig. 1. Molecular and molecular-morphological combined phylogenetic tree of the subfamily Tagiadiinae. A) Time-calibrated molecular phylogenetic tree. Numbers on the nodes indicate the molecular dates in millions of years and gray node bars represent the 95% credibility interval of divergence time estimates. B) Molecular-morphological combined phylogenetic tree. Three numbers on each node indicate JS, JG, BSV support values. C) Topology comparison of molecular tree (left panel) and molecular-morphological combined tree (right panel).

Morphological Characters and Phylogenetic Tree

A data matrix (62 morphological or biological traits $\times$ 53 hesperiid species) was obtained (Table 3). Fourteen multistate characters (character 1, 4, 7, 10, 12, 22, 25, 27, 34, 38, 47, 51, 53, and 58) were treated as additive and the other characters were treated as binary.

A strict consensus tree (tree length = 275, Consistency index = 0.26, Retention index = 0.64) was generated from a total of 50 MP trees by using TNT software (Fig. 2). Cladistic analyses recovered the monophyly of the tribe Celaenorrhinini with the inter-genus relationships (*Pseudocoladenia* + (*Sarangesa* + (*Aurivittia* + *Celaenorrhinus*))), and the intragenus relationships within *Celaenorrhinus* is that (*C. consanguineus* + *C. aspersus* + (*C. kiku* + (*C. maculosa* + (*C. patula* + *C. pulomaya*))))). In maintaining the topology of the intra-subfamily relationships of Tagiadiinae, 3 synapomorphies were observed: (i) the length of the male abdomen is shorter than or equal to the posterior margin of the hindwing (character 44: 0); (ii) during resting of adults, the wings are either spread flat or overlapping above the abdomen (character 60: 1); and (iii) when adults sunbathe, the forewings and hindwings are outspread (character 61: 1) (Fig. 2). For 3 tribes, although a presumptive synapomorphy -male forewing anterior margin fold- is found in Netrocorynini, only one representative was included in our analysis. In addition, there are only homoplastic characters supporting the monophyly of Tagiadini + Celaenorrhinini (Fig. 2).

Morphological data substantially corroborate the monophyly of the subfamily Tagiadiinae and the tribe Celaenorrhinini. However, the monophyly of the tribe Tagiadini and the intricate internal phylogenetic associations among its constituent genera continue to pose challenges for conclusive determination.

Combined Molecular-Morphological Phylogenetic Tree

A phylogenetic tree inferred from the combined data was constructed using TNT (Fig. 1B). Due to the different taxon sampling between the molecular-based and combined molecular-morphological phylogenetic trees, we compared their topologies by highlighting differences in the intra-subfamily relationships. This comparison is illustrated in Fig. 1C. The result showed that the monophyly of the 3 tribes within Tagiadiinae was well recovered in both analyses (Fig. 1A and B and Supplementary Figs. S1 and S2), with strong support for the placement of the tribe Netrocorynini as the basal lineage of Tagiadiinae. However, some notable topological differences of the tribe Tagiadini were observed between the 2 trees. For example, in the combined tree, the genus *Satarupa* was placed as a basal clade in Tagiadini, whereas it had a closer relationship with the clade consisting of (*Darpa* + *Seseria* + *Abraximorpha* + *Caprona* + *Odontoptilum* + *Abantis*) in the molecular tree. In addition, the clade consisting of (*Tapena* + *Ctenoptilum* + *Gerosis*), along with *Triskelionia* and



Fig. 2. Cladistic tree of the subfamily Tagiadinae inferred from the 62 morphological or biological traits. Black and white circles at branches represent the homologous and nonhomologous characters, respectively. The numbers above and below the circle represent the character number and its corresponding trait, respectively. The number at each node indicates BS value.

Table 3. Morphological data matrix of the subfamily Tagiadinae

Species	000000000111111111222222222333333333444444444555555555666 12345678901234567890123456789012345678901234567890123456789012
<i>Abantis efulensis</i>	200011110100010100000201112100001001010000001???011101102011?
<i>Abraximorpha davidii</i>	10001120000000010001020110010000100101010100110101101001010110
<i>Abraximorpha esta</i>	10001120000000010001020110010000100101010100110101101001010110
<i>Aurivittia aurivittata</i>	11011110010101100001021110110000100101010001002010010011000110
<i>Aurivittia vietnamicus</i>	100111100101-00000010111201100001001110-0001002010010021010110
<i>Barca bicolor</i>	110011002-10010010000101111100001001010100110??0:0000001110001
<i>Capila jayadeva</i>	10001110001000100001020011010000000100011010100000201011011110
<i>Capila neolineata</i>	10001110001000100001020010110000100100010010101000201001010110
<i>Capila pennicillatum</i>	10001110000000100001020011110000000100010010100000201001010110
<i>Capila translucida</i>	10001110001000100001020011110000000100010010100??0201011011110
<i>Caprona agama</i>	210011110001000100010201111100000101110--000111001000011010110
<i>Caprona alida</i>	21001111000100010001021121110000110112010000111001000011010110
<i>Celaenorrhinus aspersus</i>	100111?00101?11000010201101100001001020100010020100-30-1000110
<i>Celaenorrhinus consanguineus</i>	10011120010111100001010110110000100101000001002010013011000110
<i>Celaenorrhinus kiku</i>	10011120010111100001020110010000100102010001002010013011000110
<i>Celaenorrhinus maculosus</i>	10011120010111100001010010010000100101010001002010003011000110
<i>Celaenorrhinus patula</i>	100111200101111000010200110100001001010100010020100030-1000110
<i>Celaenorrhinus pulomaya</i>	100111200001111000010211110100001001010100010020100030-1000110
<i>Coladenia boenei</i>	100111111110001000010211111100001001110110101100100001020110
<i>Coladenia laxmi</i>	10001111010000100001020111110000100101111000001100100001010110
<i>Coladenia vitrea</i>	10011111111000100000010111110000100101010010101100101001020110
<i>Ctenoptilum vasava</i>	1000111101000010000002011011000012010111000000000200011010110
<i>Darpa hanria</i>	100111110101011-0001020111110000110101011100-???0103011011110
<i>Darpa pteria</i>	100111110101011-0001020111110000100101010100-???0103011011110
<i>Gerosis phisara</i>	11011111111100100001020111110000100111011001001100100011010110
<i>Gerosis sinica</i>	11011111111100100001020011010000100101010001000100100021010110
<i>Halpe nephele</i>	11011110000000000001000011110110001102001001001010111001010001
<i>Heteropterus morpheus</i>	00011111001100001110020110110000000101010001001010101001010001
<i>Mooreana trichoneura</i>	10001120000000100001020011010000000101010010111001002000011111
<i>Netrocoryne repanda</i>	101011100010000?0?0?010?0?0?10?00201?1?100000?0?0?0?0?0?0?11?
<i>Odontoptilum angulatum</i>	21101121000001010001120110110000010102010100111101000011010110
<i>Odontoptilum pygela</i>	211011210000010100011201101100000101010-01001??1?1000011010110
<i>Pintara bowringi</i>	11011110010000100001020111110000100100010000-0???0100001011110
<i>Pintara pinwilli</i>	11011110010000100001020111110000100101001000-0???0100001011110
<i>Pseudocoladenia dan</i>	11001110010001100001010111010000100101011000000000113000010110
<i>Pseudocoladenia dea</i>	10011110010001100001010110110000100101010000002010113000010110
<i>Pseudocoladenia festa</i>	10--1110010001100001020111110000100101010000001000113000010110
<i>Sarangesa dasahara</i>	10011110010001100001011110110000100100010000002000110021120110
<i>Satarupa monbeigi</i>	10001111000000100001020111010000100100110001001000100021020110
<i>Satarupa nymphalis</i>	1000111100000010000102011101000000000000101001000100021020110
<i>Satarupa valentini</i>	10001111000000100001020110010000100100001001001000100011010110
<i>Satarupa zulla</i>	11011111010000100001020111010000000100000011001000100021010110
<i>Seseria dohertyi</i>	11001111000100010000020111110000100101011100-??1?0103111010110
<i>Seseria sambara</i>	11001110000100010000021111110000100101001100-??1?01031111010110
<i>Tagiades cohaerens</i>	11011110000001100001010121110000100101010100101100000000010111
<i>Tagiades gana</i>	1101111000000100000102011111000000010101010010110010001001111-

Table 3. Continued

Species	00000000011111111122222222333333334444444455555555666 12345678901234567890123456789012345678901234567890123456789012
<i>Tagiades japetus</i>	1101111000000110000001012111000010010100100010110010001001111-
<i>Tagiades litigiosa</i>	11011110000001100001010121110000000102001100101100100000011111
<i>Tagiades menaka</i>	110111100000011000010101211100001001010100101100000000010111
<i>Tagiades nestus</i>	110111100000011000010201110100001001010001001011000000-001011-
<i>Tagiades tethys</i>	1101111000010110000101011110000100101010010101100100000010111
<i>Tagiades trebellius</i>	1101111000000110000102011101000010010200110010110010000001111-
<i>Tapena thwaitesi</i>	11--1110011000100001020120010000120111010000????010101101011?

Notably, format symbols = "0123", missing =?, gap = -.

Eagris, is placed at the basal position of Tagiadini in the molecular tree. In contrast, this clade is positioned as a later-diverged lineage in the combined tree.

Diversified Time and Biogeographic Region of Tagiadinae

Divergence time estimation showed that the crown age of Tagiadinae is approximately 37 Ma (Fig. 1A). The 2 tribes Celaenorrhini and Tagiadini were diverged approximately 30.1 Ma and 30.9 Ma, respectively. The divergence times were more recent than the estimates of Kawahara et al. (2023). The results of ancestral region reconstruction showed that the most recent common ancestor of Tagiadinae was distributed in the Indo-Malayan region (Fig. 3). In addition, a total of 69 dispersal events and 16 vicariance event were identified across the Tagiadinae lineages (Fig. 3).

Discussion

In our phylogenetic analyses, the monophyly of the internal 3 tribes of this subfamily are strongly supported (support values > 95%, Supplementary Figs. S1 and S2), with the tribal relationships (Netrocorynini + (Celaenorrhini + Tagiadini)); however, the relationships are inconsistent with the phylogenetic consensus (Warren et al. 2009, Sahoo et al. 2016, Li et al. 2019, Xiao et al. 2022, Kawahara et al. 2023). The genus *Netrocoryne* C. & R. Felder (1867) has long been placed at the basal position within the tribe Tagiadini (Warren et al. 2009, Sahoo et al. 2016), Li et al. (2019) established the tribal rank, Netrocorynini, for *Netrocoryne*, although Li et al. also obtained consistent topologies with the former 2 studies. In this study, we prefer to put *Netrocoryne* species as the basal lineage of the whole subfamily and be sister groups to the remaining Tagiadinae species whether it is morphological cladistics or molecular phylogenetic analyses (UFB/BPP/JS/JG/BSV = 92/0.985/98/100/12, Fig. 1A and B and Supplementary Figs. S1 and S2). This means that if the genus *Netrocoryne* is traditionally recognized to belong to Tagiadini *sensu* Warren et al. (2009) and Sahoo et al. (2016), the monophyly of this tribe would not be recovered. Although the relationships we obtained were different from most previous studies (Warren et al. 2009, Sahoo et al. 2016, Li et al. 2019, Xiao et al. 2022, Kawahara et al. 2023), it seems to provide the novel morphological- and molecular-based evidence for the raising of a tribe rank.

For genus level, our molecular dataset based on the expanded taxon sampling contains 30 out of 35 currently described genera of Tagiadinae representing a relatively comprehensive evolutionary framework of this subfamily. The general inter-genus relationships

are Netrocorynini: ((*Netrocoryne* + *Exometoeca*) + *Chaetocneme*), Celaenorrhini: ((*Apallaga* + *Aurivittia*) + ((*Sarangesa* + *Pseudocoladenia*) + *Colaenorrhinus*)) and Tagiadini: ((*Eagris* + (*Triskelionia* + ((*Tapena* + *Ctenoptilum*) + *Gerosis*))) + (((*Procampa* + (*Calleagris* + *Satarupa*)) + ((*Odina* + *Darpa*) + (*Seseria* + (*Abraximorpha* + ((*Caprona* + *Odontoptilum*) + (*Leucochitonea* + (*Netrobalane* + *Abantis*)))))) + (*Tagiades* + (*Mooreana* + *Pintara*) + (*Coladenia* + *Capila*))).

Several previous phylogenetic studies explored the intra-relationships of Tagiadinae (Warren et al. 2009, Sahoo et al. 2016, Toussaint et al. 2018, Li et al. 2019, Kawahara et al. 2023). However, the limited taxon sampling of these studies largely restricted exploration of phylogenetic relationships. Notably, Zhang et al. (2022) provided a relatively comprehensive evolutionary framework of Tagiadinae, which differed from our analyses in some major clades. Specifically, within Tagiadini, (Zhang et al. 2022) put the clade consisting of the 4 genera *Capila* + *Coladenia* + *Pintara* + *Mooreana* at the basal position of Tagiadini, whereas we put them at the tip position in both molecular and molecular-morphological combined analyses (Fig. 1A and B). In addition, we think these 4 genera are closely related to the genus *Tagiades*, which is supported by both molecular and molecular-morphological data, whereas Zhang et al. (2022) placed *Tagiades* to the clade consisting of (*Eagris* + *Triskelionia* + *Gerosis* + *Tapena* + *Calleagris*). In addition, relationships of certain genera and their closely related taxa also showed inconsistencies between the 2 works, for example, *Eagris* and *Calleagris*. However, Zhang et al. (2022) suggested to move *Odontoptilum* and 2 *Caprona* species (*C. alida* and *C. agama*) into genus *Abaratha*. Though our molecular, molecular-morphological combined and morphological trees also support the monophyly of the clade formed by these 2 genera, no morphological synapomorphies can be identified to substantiate this reclassification. Therefore, 2 hypotheses remain equally acceptable: (i) the perspective of Zhang et al. (2022), and (ii) treating them as independent genera. Furthermore, another study by Shen et al. (2022) focused on 42 Chinese species across 18 genera of related Tagiadini taxa based on 5 fragments of 4 mitochondrial genes (*COI*, *COII*, 28S *rDNA*, and 18S *rDNA*) showed that the internal relationships of this tribe were also still inconsistent with both Zhang et al. (2022) and ours. Within the tribe Celaenorrhini, the inter-genus relationships were consistent with the recent studies.

It is worth noting that although our analyses included a molecular-morphological combined matrix, its taxon sampling differs from that of the molecular-only dataset, being a subset of the latter. This discrepancy in taxon sampling may inherently lead to inconsistencies in phylogenetic inference. In addition, the high level of missing data for certain species in the

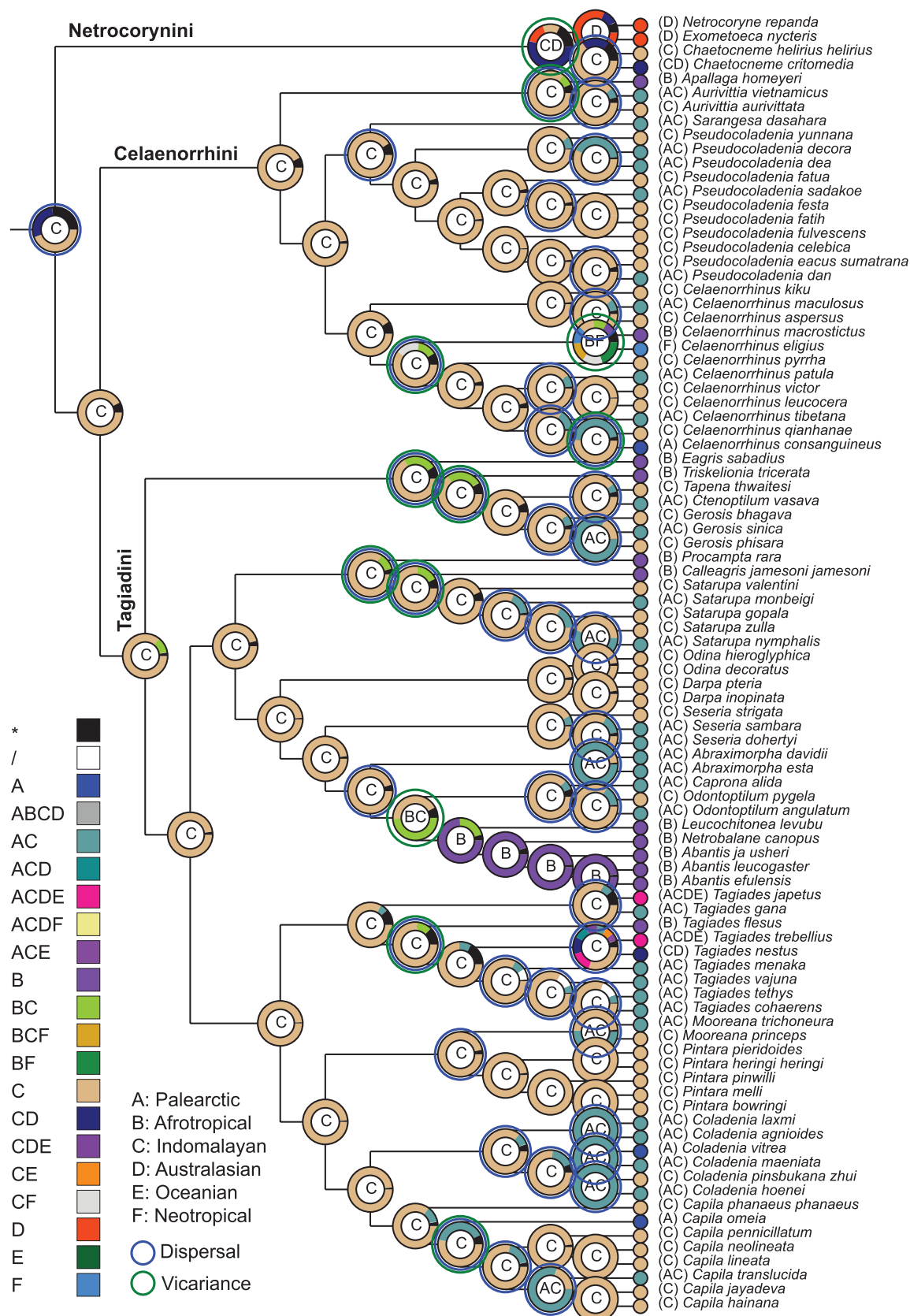


Fig. 3. Biogeographic history of the subfamily Tagiadinae. The letter at nodes indicates the predicted distribution of the most recent common ancestor. The blue and green circles indicate the dispersal and vicariance events, respectively.

morphological matrix, such as *Netrocoryne repanda*, may potentially mislead the phylogenetic analysis. The morphological tree was constructed to serve as a framework to assess the presence of synapomorphies within specific lineages. Overall, this study integrates molecular, morphological, and combined phylogenetic analyses to provide a comprehensive framework for understanding the evolutionary relationships within Tagiadinae and evaluating the influence of morphological evidence in phylogenetic inference.

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Author contributions

Xiangyu Hao (Conceptualization [equal], Formal analysis [equal], Investigation [equal], Methodology [equal], Validation [equal], Writing—original draft [equal]), Yue Pan (Conceptualization [equal], Data curation [equal], Formal analysis [equal], Investigation [equal], Supervision [equal], Writing—original draft [equal]), Hideyuki Chiba (Conceptualization [equal], Validation [equal], Writing—review & editing [equal]), and Xiangqun Yuan (Conceptualization [equal], Funding acquisition [equal], Project administration [equal], Resources [equal], Writing—review & editing [equal])

Supplementary material

Supplementary material is available at *Insect Systematics and Diversity* online.

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Conflicts of interest. None declared.

Data availability

All the nucleotide sequences of 3 genes COI, COII, and EF-1a amplified by PCRs are deposited in GenBank of NCBI with accession numbers PP693152-PP693205 and PP707091-PP707191. The matrix used for phylogenetic reconstruction and treefiles were deposited in FigShare (<https://doi.org/10.6084/m9.figshare.26411164>).

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